

Deterministic pose-spline dynamics estimator

Estimator structure

pose-only continuous-time spline -> analytic derivatives
pose spline degree: 5 (position and normalized quaternion)
fixed-spline gradient matching -> shared physical parameters and lag
default physical initialization: exact nominal chart origin
previous estimator result loading: explicit opt-in only
shooting nodes: none
continuity constraints: none
sensor/IMU channels in loss: none

Initial estimate

source kind: nominal
source path: None
source mass [kg]: 2.35159759081
selected initial mass [kg]: 2.35159759081

Selected strict-ZOH parameters

mass [kg]: 2.77007067565
inertia [kg m²]:
[0.0723680002638, -0.00215212693168, 6.60512665078e-05]
[-0.00215212693168, 0.154157902784, 0.000160004678748]
[6.60512665078e-05, 0.000160004678748, 0.225561277782]
principal moments [kg m²]: [0.0723113785965, 0.154214141423, 0.22556166081]
inertia triangle margin [kg m²]: 0.000963859209661
CoG offset [m]: [-0.00219756920562, 0.000734432188529, -0.0220752326934]
rotor force effectiveness: [0.99185433828, 1.04452447908, 0.978566891551, 0.986377057864]
lag [s]: 0.038868240958
joint dynamics loss: 0.178477846339
soft-prior cost: 0.0531116909237

Smooth physical coordinates

log_mass_scale	0.163777911491
log_second_moment_cholesky_xx_scale	0.434303665249
log_second_moment_cholesky_yy_scale	0.0538431659632
log_second_moment_cholesky_zz_scale	0.00169823212153
normalized_second_moment_cholesky_yx_offset	0.0216058838445
normalized_second_moment_cholesky_zx_offset	-0.00125479552757
normalized_second_moment_cholesky_zy_offset	-0.00795085414978
cog_offset_x_m	-0.00219756920562
cog_offset_y_m	0.000734432188529
cog_offset_z_m	-0.0220752326934
force_effectiveness_contrast_1	-0.0365862399948
force_effectiveness_contrast_2	0.0321352582
force_effectiveness_contrast_3	0.0158385490699

Spline and collocation

spline degree: 5
collocation step [s]: 0.01

Bag failure_1

selected spline knot spacing [s]: 0.2
spline fit interval [s]: [19.0092086792, 24.9592086792]
parameter-estimation interval [s]: [19.6042086792, 24.3642086792]

excluded knot spans at each spline boundary: 3
actual boundary exclusion [s], start/end: [0.595, 0.595]
dynamics loss: 0.274763331312
estimated forward position RMSE [m]: 50.7212899719
estimated + external wrench position RMSE [m]: 74.428025573
external wrench improves estimated free rollout: False
nominal forward position RMSE [m]: 61.968178935
estimated improves nominal: True
residual wrench mean: [0.015983 0.047153 0.114443 -0.037234 0.038256 -0.006387]
residual wrench RMSE: [0.557345 0.306524 1.19605 0.069361 0.479492 0.134936]
residual wrench trend/s: [-0.094774 0.054124 -0.535051 -0.011014 0.086096 0.00717]
residual wrench cumulative impulse: [0.069929 0.22393 0.550073 -0.177788 0.182601
-0.028672]

Bag failure_2

selected spline knot spacing [s]: 0.2
spline fit interval [s]: [25.5140545368, 30.9640545368]
parameter-estimation interval [s]: [26.0979831082, 30.3779831082]
excluded knot spans at each spline boundary: 3
actual boundary exclusion [s], start/end: [0.583928571429, 0.586071428571]
dynamics loss: 0.082192361367
estimated forward position RMSE [m]: 53.9933985912
estimated + external wrench position RMSE [m]: 31.7452341114
external wrench improves estimated free rollout: True
nominal forward position RMSE [m]: 66.5168930072
estimated improves nominal: True
residual wrench mean: [0.100267 -0.324288 -0.121313 0.037526 -0.010797 0.017059]
residual wrench RMSE: [0.212873 0.383962 0.671271 0.054906 0.221848 0.084244]
residual wrench trend/s: [-0.053959 0.057419 -0.448224 -0.012237 0.00941 0.003789]
residual wrench cumulative impulse: [0.428758 -1.387366 -0.523687 0.160717 -0.048483
0.073007]

Smoothstep continuation

width=0.5: delay=0.0381215801s, cost=0.231441860644, nfev=14
width=0.2: delay=0.037868071s, cost=0.23158612457, nfev=8
width=0.05: delay=0.037868241s, cost=0.231679846718, nfev=2

Strict-ZOH polish

delay=0.033868241s: screening=0.233717531653, refined=nan, selected=False
delay=0.034868241s: screening=0.233703270236, refined=nan, selected=False
delay=0.035868241s: screening=0.233683558728, refined=nan, selected=False
delay=0.036868241s: screening=0.233640932667, refined=nan, selected=False
delay=0.037868241s: screening=0.231770490005, refined=nan, selected=False
delay=0.038868241s: screening=0.231589598614, refined=0.231589537263, selected=True
delay=0.039868241s: screening=0.231646587259, refined=0.231646506578, selected=False
delay=0.040868241s: screening=0.231639092216, refined=0.231638779725, selected=False
delay=0.041868241s: screening=0.231663602203, refined=nan, selected=False